

Report - Version 8: Autonomous Experiment Planning

Version 8 extends **Building Agentic AutoML** with a **Senior Agent** responsible for autonomous experiment planning.

The Version 7 search space remains available, but it is no longer explored exhaustively. The Senior Agent uses dataset inspection and previous results to choose which experiment to run next, under a fixed budget and with a transparent decision history.

With 2 preprocessing strategies, 2 feature engineering strategies, 3 models, and 2 configurations, **24 candidate experiments** remain available. With the default budget of 10, the agent executes **10 experiments** (58.33% reduction) and at most 50 fits with 5-fold cross-validation, instead of the 120 fits required by Version 7 exhaustive search.

Goal of Version 8

- preserve the components introduced in Versions 1-7;
- distinguish binary classification, multiclass classification, and regression;
- build the same 24-candidate pool while executing only useful experiments within the budget;
- establish baselines for LightGBM, XGBoost, and CatBoost and adapt later choices to observed results;
- record decision reason, status, score, execution time, and best-score trajectory;
- return the best observed configuration together with the complete decision history.

Architecture

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Dataset + Target -> Task/Features -> Data Inspection -> Candidate Experiment Space  
-> Senior Agent Planner -> Selected Experiment -> Cross-Validation Evaluation  
-> Decision History / State -> Best Observed Configuration
```

Task Detection

Version 8 extends task detection to distinguish binary classification, multiclass classification, and regression. Non-numerical targets are treated as classification; numerical targets use both the number of unique values and their ratio relative to dataset size.

Feature Detection

Features are still split into numerical and categorical groups from dataframe dtypes. On Adult Income, the agent detects 6 numerical features and 8 categorical features.

Data Inspection

Inspection now stores dataset shape, target type and distribution, dtypes, missing counts and percentages, cardinality, and duplicate rows. On Adult Income it reports 48,842 rows, 15 columns, 52 duplicates, and missing values in workclass = 2,799, occupation = 2,809, and native_country = 857.

Preprocessing

The **native** and **impute** strategies remain available. Imputation parameters are learned exclusively from the training fold, with more robust handling of completely missing columns.

Feature Engineering

The **none** and **interactions** strategies remain available. Interactions are pairwise products between numerical features and are created inside each fold after preprocessing.

Model Selection

LightGBM, XGBoost, and CatBoost remain the three candidates. Classifiers are initialized consistently for binary or multiclass objectives; regression models remain unchanged.

Experiment Space

The combinations define a pool of **24 candidates** (2 preprocessing x 2 feature engineering x 3 models x 2 configurations), separating the available search space from the policy that decides what to execute.

Metric Selection

ROC AUC for binary classification, **ROC AUC OvR Macro** for multiclass classification, and **RMSE** for regression.

Smart Validation

StratifiedKFold is used for classification and **KFold** for regression. The default is 5 folds; for classification, the number of folds is reduced automatically when the smallest class contains fewer than 5 observations.

Training and Evaluation

Each selected candidate is evaluated across folds with preprocessing and feature engineering kept inside validation. Version 8 adds multiclass support, execution-time tracking, explicit status, and failure handling without stopping the entire AutoML process.

Senior Agent Planning

This is the new capability introduced in Version 8. The policy is deterministic and sequential: it uses completed results and missing-value information to select the next experiment.

- Stage 1: establish a native + none + default baseline for every model family;
- Stage 2: try the alternative hyperparameter configuration of the current best model;
- Stage 3: when missing values exist, compare alternative preprocessing while keeping the rest of the best configuration fixed;
- Stage 4: test numerical interactions around the current best configuration;
- Stage 5: use the remaining budget to explore untested candidates.

Results - Adult Income

Cfg: **B** = baseline; **T** = alternative configuration. The planner executes 10 experiments out of 24.

Step	Preproc.	Feature Eng.	Model	Cfg	Mean AUC	Sec
1	native	none	LightGBM	B	0.929556	4.08
2	native	none	XGBoost	B	0.926851	4.44
3	native	none	CatBoost	B	0.930631	250.66
4	native	none	CatBoost	T	0.929910	121.42
5	impute	none	CatBoost	B	0.930119	198.22
6	native	interactions	CatBoost	B	0.929786	215.24
7	native	none	LightGBM	T	0.929846	4.57
8	native	none	XGBoost	T	0.926509	4.43
9	native	interactions	LightGBM	B	0.928578	3.49
10	native	interactions	LightGBM	T	0.928841	5.91

Best observed configuration: **CatBoost + native + none + baseline**, mean ROC AUC = **0.9306306263**, std = **0.0022463105**.

Budget: **10/24 experiments** - reduction = **58.33%** - observed total duration = **812.45 s**.

Decision History

The first three steps establish the model baselines. CatBoost immediately becomes the best model; tuning, imputation, and interactions do not beat the baseline on Adult. The remaining steps use the residual budget for exploration. The Version 7 best solution is therefore recovered without executing all 24 candidates.

Generalization Tests

The same agent() function is executed without changing the planning policy on Wine, mixed-type California Housing, and numeric California Housing. Every benchmark uses 10 experiments out of 24 and 5-fold cross-validation.

Dataset	Task	Metric	Best	Preproc.	Feature Eng.	Score	Std	Sec
Adult Income	binary	ROC AUC	CatBoost	native	none	0.930631	0.002246	812.45
Wine	multiclass	AUC OvR Macro	CatBoost	native	none	0.998847	0.001814	90.39
California Housing	regression	RMSE	CatBoost	impute	interactions	45482.05	1289.98	229.36
California Numeric	regression	RMSE	CatBoost	native	interactions	45703.19	1375.85	86.30

Wine - Multiclass Classification

The task is detected as multiclass, the metric becomes ROC AUC OvR Macro, and validation remains StratifiedKFold. CatBoost + native + none reaches 0.998847 with std 0.001814. With no missing values, imputation is not prioritized in the 10 observed steps.

California Housing - Mixed-Type Regression

The task automatically switches to regression with KFold and RMSE. Missing total_bedrooms values trigger the alternative preprocessing comparison: the best observed configuration becomes CatBoost + impute + interactions, RMSE 45,482.05.

California Housing - Numeric Regression

The complete numeric dataset has no missing values. The policy spends its budget on models, tuning, and feature engineering; the best result is CatBoost + native + interactions, RMSE 45,703.19.

Planner Behavior

The results show that the same policy follows different trajectories according to the data: on mixed California Housing it explores imputation because missing values are present, while on the numeric representation it does not prioritize imputation. The decision history makes this difference explicit.

What Version 8 Teaches

Version 8 moves the project from exhaustive search to adaptive search: the available tools are preserved, but a new component decides which experiments deserve the available budget.

On Adult Income, the Senior Agent recovers the **same best configuration as Version 7** (CatBoost + native + none + baseline) while executing 10 experiments instead of 24. This demonstrates the value of planning without claiming that the rule-based policy guarantees a global optimum.

The evolutionary pattern remains explicit: V1 builds the baseline, V2 observes the data, V3 experiments with preprocessing, V4 compares models, V5 makes validation more robust, V6 compares feature spaces, V7 compares hyperparameter configurations, and V8 autonomously decides which experiments to execute. The One Version, One Concept principle remains intact.

Limitations

- the planning policy is intentionally simple, deterministic, and rule-based;
- the budget is expressed as experiment count; execution time is recorded but is not yet used to guide choices;
- no Bayesian optimization, reinforcement learning, LLM planning, or dedicated early stopping;
- the best configuration is selected and reported from the same cross-validation process used for model selection, with no independent final holdout;
- task detection remains heuristic for discrete numerical targets;
- feature engineering is limited to pairwise numerical interactions and only three boosting model families are available;
- the state does not yet store a persistent fitted pipeline.

Conclusion

Version 8 introduces Autonomous Experiment Planning without replacing the components built in previous versions. The Senior Agent builds the candidate space, plans and executes a limited sequence of experiments, records the reason behind every choice, and returns the best observed configuration with the complete decision history. The project therefore moves from exhaustive AutoML search to a first adaptive, transparent, budget-aware agent.